

Amr Mahmoud | Software Engineer | Backend Engineer

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Summary

Back-End Engineer and Computer Science student with experience building RESTful APIs, authentication systems, and database-driven backends using Node.js, Laravel, and Flask. Strong foundation in data structures, system design, and Linux-based development environments.

Experience

CS-Club Vice-President & Instructor: *E-JUST Clubs* February 2026 – Present

- Designed and delivered a 3-lecture Version Control curriculum (**Git & GitHub**), enabling students to manage codebases and collaborate effectively via pull requests and branching workflows.

Android Development Trainee: *Digital Egypt Pioneers Initiative (DEPI)* July 2025 – November 2025

- Built native Android applications using **Kotlin** and **Jetpack Compose**, following **MVVM** architecture with clean separation of UI and business logic.

IT Field Intern: *Egypt Japan University of Science and Technology (E-JUST)* July 2025 – September 2025

- Developed and maintained backend features for the university website and course catalogue using **PHP Laravel**.
- Built the Student Affairs portal with **Laravel & React**, integrating **Paymob** payment gateway for online transactions.
- Applied networking fundamentals (TCP/IP, routing, security) in a real infrastructure environment.

Education

Bachelor of Science in Computer Science and Information Technology

Egypt Japan University of Science and Technology (E-JUST) 2023 – 2027 (Expected)

- GPA: 3.51/4.0
- Current Level: 4th Year
- Full Scholarship Recipient – Innovators Support Fund (**ISF**)
- CS-Club Vice President

Projects

Light Sub Pro – Subscription Management Telegram Bot (@light_sub_pro_bot):

- Engineered a production-ready Telegram bot handling the full subscription order lifecycle - from product discovery and payment to fulfillment and Telegram-based admin management.
- Integrated **4 payment gateways** (InstaPay, Vodafone Cash, Bybit, Binance) with automated crypto verification via Bybit/Binance APIs and SMS detection via **SMSGate** app; built a **merchant reseller system**, **promo code engine**, and **in-bot wallet**.
- Delivered **i18n**, broadcast notifications, and a **Hono REST API** for external integrations - all backed by optimised **PostgreSQL** queries with no caching layer.
- Technologies:** Bun, Hono, PostgreSQL, grammY, TypeScript, Railway

Green Rabbit:

- Integrated third-party AI services via the Perplexity AI API to power an intelligent chatbot.
- Secured API endpoints with JWT authentication and applied rate limiting through Redis
- Documented the RESTful interfaces using API Dog to streamline frontend integration.
- Technologies:** Node.js, Express.js, PostgreSQL, JWT, API Dog, BullMQ, Perplexity AI, cron jobs, Railway

Campus Connect: GitHub

- Developed a cross-platform university engagement portal for E-JUST students featuring event management, club discovery, room booking, facility reservations, and social feed with posts and interactions.
- Designed relational database schema and implemented secure authentication and authorization flows used by web and mobile clients.
- Built RESTful API with Express.js implementing JWT authentication, (RBAC) Role-Based Access Control, and layered architecture (Controllers, Services, Repositories) connected to MariaDB.
- Technologies:** Node.js, Express.js, MariaDB, zod JWT, API Dog, PM2

Skills

Programming Languages: TypeScript, JavaScript, Python, PHP, Java, Kotlin, C++

Backend & Frameworks: Node.js, Express.js, Hono, Bun, Laravel, Flask, grammY

Databases & Tools: PostgreSQL, MariaDB, Redis, Docker, Git, Linux, API Dog, Railway, Cloudflare

Concepts: REST API Design, JWT & RBAC, OOP, Data Structures & Algorithms, System Design

Languages: Arabic (Native), English (Upper Intermediate)